FORM FOUR MOCK EXAMS 2024

CHEMISTRY 1 MARKING SCHEME

01.

i	ii	iii	iv	v	vi	vii	viii	ix	x
C	D	A	B	B	C	A	B	B	C

10marks

02

i	ii	iii	iv	v	vi
C	D	A	B	E	B

06Marks

SECTION B

03 (a) A - Experimentation 01

B. Conclussion 01

C. Reporting the results. 01

(b)(i) Dependent variable - The solubility of ammonium chloride because is a factor unde investigation 02

(ii) Independent variable - The temperature of water since it varies 02

(iii) Controlled variable - The amount of water because is kept constant. 02

04 Methods of removing hardness of Water:

a) Boiling
b) The use of washing soda
c) Ion exchange
d) Distillation

Any 03 Points (03 @ = 09 marks)

05 (a) (i) Empirical formula is the simplest formula which represents the simplest ratio of atoms in a compound or molecule. (01 mark)

(ii) Molecular formula is a formula which represents the actual number of atoms present in a compound or molecule. (01 mark)

(b)

Element	C	H	O
Percentage composition	52·2	13	34·8
Divide by Relative Atomic Mass	$\frac{52·2}{12}$	$\frac{13}{1}$	$\frac{34·8}{16}$
	=4·35	=13	2·175
Divide by the Smallest Value throughout	$\frac{4·35}{2·175}$	$\frac{13}{2·175}$	$\frac{2·175}{2·175}$
	= 2	= 5·977	= 1
Ratios	2	≈ 6	1

(00½ marks @ stage = 03 Marks)

∴ The Empirical formula = C_2H_6O (00 1 mark)

But

Empirical formula Mass = n x Molecular Mass

Molecular Mass = 2 x Vapour density (00 1 mark)

Empirical formula Mass = n x 2(Vapour density)

$46 = n \times 2 \times 23$

$46 = 46n$

$n = \underline{1}.$ (001 mark)

Then:

Molecular formula = n.x Empirical formula

Molecular formula $= (C_2H_6O)\underline{1}$

$= C_2H_6O.$

∴ The Molecular formula of the compound is C_2H_6O (01mark)

06. (a) Pipette 01½

(b) Colourless 01½

(c) Acid which is Hydrochloric acid was more concentrated than the other because only 22cm³ neutralized 25cm³ of base 02

(d) The salt formed is Sodium Chloride (NaCl) 02

(e) To obtain pure crystals of the salt we do Evaporation / simple distillation. 02

07. (a) Sodium Chloride (solid Chloride). 02 marks

(b) The gas C is Hydrogen chloride gas 02 Marks

(c) White ppt is Silver Chloride. 02 Marks

(d) Ionic equation

$$Ag^{+}_{(aq)} + Cl^{-}_{(aq)} \longrightarrow AgCl_{(s)}.$$ 03 Marks.

08. (a) The reaction is Exothermic reaction 01 Mark

(b) Application of Exothermic reaction

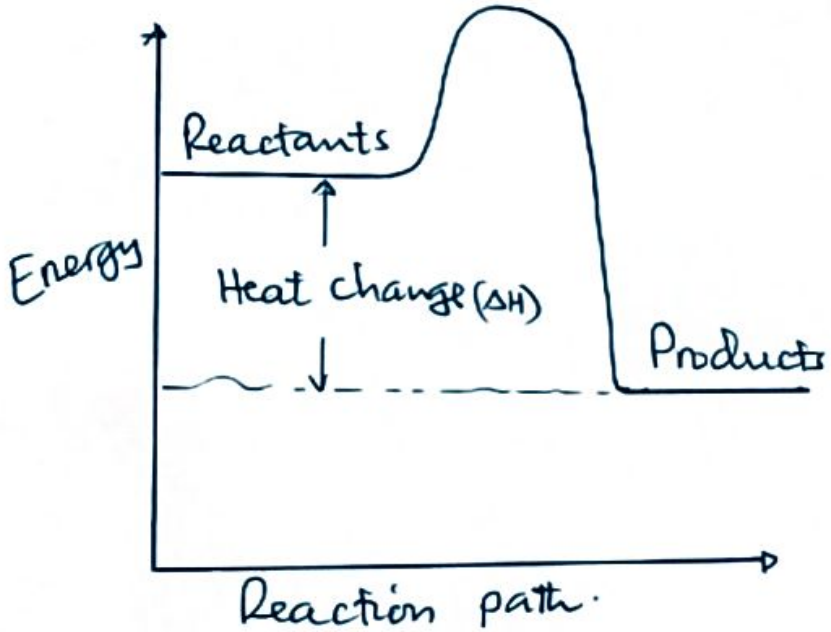
(i) Combustion in engines 01

(ii) Helps in burning of candles 01

(iii) Cooking at home as heat is given out. 01

Any 3points = 03marks.

08 (c) Energy level profile for exothermic reaction.



H = 01
L = 02
D = 02

SECTION C.

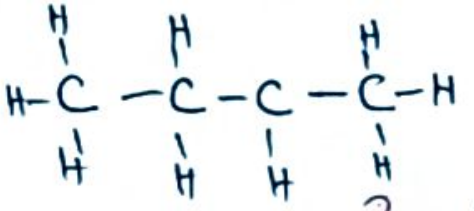
09 (a)(i) Functional group is a group of atoms with a characteristic chemical reactivity. 01mark

(ii) Homologous series is a series of organic compounds with the same functional group and similar chemical properties in which the members differ from each other by $-CH_2$ group. 01mark

(iii) Isomerism - is the existence of organic compounds having the same chemical formular but different structural formular. 01mark.

09 (b) Isomers of C_4H_{10}

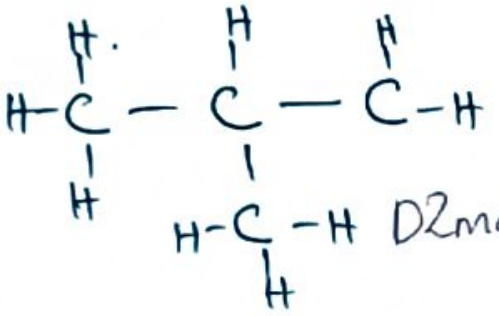
(i) Butane $CH_3CH_2CH_2CH_3$



02mark

(ii) 2-methylpropane

$CH_3CH(CH_3)CH_3$



02mark.

(c)(i) Ethene to ethane.

$$CH_2=CH_{2(g)} + H_{2(g)} \xrightarrow[130^\circ C]{\text{nickel}} CH_3CH_{3(g)}.$$ 02 Mark

(ii) Ethene to chloroethane.

$$CH_2=CH_{2(g)} + HCl_{(g)} \longrightarrow CH_3CH_2Cl_{(l)}$$ 02Mark

(iii) Ethene to 1,2-dibromoethane.

$$CH_2=CH_{2(g)} + Br_{2(g)} \longrightarrow CH_2BrCH_2Br_{(g)}$$

02Mark

(iv) Ethene to Ethanol

$$CH_2=CH_{2(g)} + H_2O_{(l)} \xrightarrow{H_2SO_4} CH_3CH_2OH_{(g)}.$$

02Mark.

10. a) (i) Equation for the reaction:

$$Zn_{(s)} + 2AgNO_{3(aq)} \longrightarrow Zn(NO_3)_2 + 2Ag_{(s)}$$ (02 marks)

(ii) Mass of Zinc Metal used is 1.63g

$$Zn_{(s)} + 2AgNO_{3(aq)} \longrightarrow Zn(NO_3)_2 + 2Ag_{(s)}$$ 01

65g of Zn ⟶ 108g of Ag

1.63g of Zn ⟶ x (X) 01

$$x = \frac{1.63g \times 108g}{65}$$

$$= 5.42g.$$ 01

∴ The Mass of Silver formed = 5.42g (01 mark)

b) (i) The Species that carried the current through the solutions are ions.

- Cations : Ag^+ and H^+ - (01 Marks)
- Anions : OH^- and NO_3^-
- Direction of Movement
- Cations which migrate to Cathode are Ag^+ and H^+ (01 marks)

Anions which move to the Anode are OH^- and NO_3^- (01 marks)

ii) Equation for the reaction took place :

At the Anode

OH^- will be discharged in preference to NO_3^-, because OH^- is lower in the electrochemical series than NO_3^-. Therefore Water and Oxygen gas will be formed at the ~~Cathode~~ Anode.

$$4OH^- \longrightarrow 2H_2O + O_2 + 4e^-$$ (01 marks)

10 b) (ii). At the Cathode

$$Ag^+ + e^- \longrightarrow Ag_{(s)}$$ (01mark)

Overall equation of the cell.

$$4OH^- \longrightarrow 2H_2O + O_2$$

$$Ag^+ + 4e \longrightarrow 4Ag^*$$

$$4OH^-_{(aq)} + 4Ag^+_{(aq)} \longrightarrow 2H_2O_{(l)} + O_2 + 4Ag_{(s)}$$ (01mark)

(iii) If platinum anode was replaced by Silver anode, the Silver anode would be oxidised and go into the solution, hence the Anode would decrease in mass. None of the ions would be discharged at the anode (01mark)

$$Ag_{(s)} \longrightarrow Ag^+ + e^-$$ (00½ mark)

iv) Data given.

Current (I) = 1.5A

Time t = 1hour and 10minutes (00½ marks)

= 60 x 60 + (10 x 60)

= 3600 + 600

= 4200sec.

From Q = It (00½ marks)

= 1.5A x 4200sec (01 marks)

= 6300 Coulombs

∴ The quantity of electricity passed (Q) was 6300C (00½ mark)

11 (a) A carbonate is not a base from the meaning of a base that is an oxide or hydroxide of a metal. Or Is a compound which ionizes to form hydroxide as the only negative ion in solution. 02marks

(b) Data:
(i) Concentration of Hydroxide $\frac{0.074g}{0.25dm^3}$

$= 0.296 g/dm^3$ 01mark.

Molarity of acid (Ma) $= n/V$

$n = \frac{\text{Volume given}}{GMV}$

$= \frac{0.1792}{22.4}$ 01mark

$= 0.008$ moles in 1 litre.

$\therefore Ma = 0.008M$

$Mb = ?$

$Va = 20cm^3$

$Vb = 20cm^3$

$na = 2$

$nb = 1.$

from

$\frac{MaVa}{MbVb} = \frac{na}{nb}$ 01mark

$Mb = \frac{MaVanb}{Vbna}$

$= \frac{0.008M \times 20cm^3 \times 1}{20 \times 2}$

$\therefore$ The molarity of $X(OH)_2$ is 0.004M 02marks

(ii) Molar mass of $X(OH)_2$

from molarity = $\frac{\text{Concentration}}{\text{Molar mass}}$.

Molar mass = $\frac{\text{Concentration}}{\text{Molarity}}$ 01mark

$= \frac{0.074g}{0.25dm^3}$

$= 0.296g/dm^3$ 01mark

$\frac{0.296g/dm^3}{0.004Mol/dm^3}$

$= 74g/mol$.

∴ Molar mass of $X(OH) = 74g/mol$. 02marks

(iv) Atomic mass x.

$X(OH)_2 = 74$ 01mark

$X + 17X2 = 74$

$X + 34 = 74 - 34$

$X = 40$

∴ The atomic mass X is 40g/mol. 02marks

(v) Element X is calcium. 01mark